MCA – Semester III

Computer Networking Skill Development

Course Code: MCA03EL04CL

Practical Journal – Aim, Objectives, Program and Conclusion (Assignments 1 to 10)

Item	Details
Program	MCA
Semester	III
Credits	1 (Practical: 2 hrs/week)
Evaluation	Term Work: 50 Marks
Tools	Cisco Packet Tracer / GNS3, Wireshark, LAN tester, crimping tool

Note: IP addresses, device names and topology below are sample values. Change them to match your own lab setup and screenshots before submitting.

Practical No. 1

Basic network devices, transmission media, RJ-45 and cable crimping

Aim

To study basic network devices and transmission media, and to prepare Straight-through and Crossover Ethernet cables using RJ-45 connectors and test them with a LAN tester.

Objectives

- To identify network devices such as hub, switch, router, NIC, modem and access point.
- To understand transmission media: UTP, STP, coaxial and fibre optic cable.
- To learn T568A and T568B wiring standards.
- To crimp RJ-45 connectors and test cable continuity.

Program / Procedure

Tools required: UTP Cat5e/Cat6 cable, RJ-45 connectors, crimping tool, wire stripper/cutter, LAN cable tester.

Colour codes:

Pin	T568A	T568B
1	White/Green	White/Orange
2	Green	Orange
3	White/Orange	White/Green
4	Blue	Blue
5	White/Blue	White/Blue
6	Orange	Green
7	White/Brown	White/Brown
8	Brown	Brown

Steps:

1. Cut a suitable length of UTP cable and strip about 1.5 inch of the outer jacket.
2. Untwist the four pairs and arrange the wires in the required colour order.
3. Straight-through cable: use T568B at both ends. Crossover cable: T568A at one end and T568B at the other.
4. Trim the wires evenly, insert them into the RJ-45 connector (clip facing down) until they reach the end.
5. Place the connector in the crimping tool and press firmly.

6. Repeat on the other end, then insert both ends into the LAN tester and check that all 8 LEDs glow in sequence (1-1 to 8-8 for straight, crossed order for crossover).

Conclusion

Straight-through and crossover cables were successfully prepared and tested. Straight-through cables connect unlike devices (PC to switch) and crossover cables connect like devices (PC to PC). The LAN tester confirmed continuity of all eight wires.

Practical No. 2

IPv4 addressing, subnetting and network commands

Aim

To configure IPv4 addresses, subnet masks and default gateway, perform subnetting, and verify connectivity using ping, ipconfig/ifconfig, tracert/traceroute and arp.

Objectives

- To understand IPv4 address classes, subnet mask and default gateway.
- To divide a network into subnets using subnet masks.
- To use network diagnostic commands to verify and troubleshoot connectivity.

Program / Procedure

Subnetting example: network 192.168.10.0/24 divided into 4 subnets (borrow 2 bits, new mask /26 = 255.255.255.192, 62 usable hosts per subnet).

Subnet	Network ID	First Host	Last Host	Broadcast
1	192.168.10.0	192.168.10.1	192.168.10.62	192.168.10.63
2	192.168.10.64	192.168.10.65	192.168.10.126	192.168.10.127
3	192.168.10.128	192.168.10.129	192.168.10.190	192.168.10.191
4	192.168.10.192	192.168.10.193	192.168.10.254	192.168.10.255

Steps:

1. Assign IP address, subnet mask and default gateway to each PC (for example PC1: 192.168.10.2 / 255.255.255.192 / gateway 192.168.10.1).
2. Open the command prompt and run the commands below.

```
ipconfig /all        (Linux: ifconfig or ip addr)
ping 192.168.10.3
ping 192.168.10.1
tracert 192.168.10.65     (Linux: traceroute 192.168.10.65)
arp -a
```

Observe the reply, TTL, route hops and the IP-to-MAC mapping in the ARP table.

Conclusion

IPv4 addressing and subnetting were performed successfully. Devices in the same subnet communicated directly, and the ping, tracert, ipconfig and arp commands helped verify and troubleshoot connectivity.

Practical No. 3

Design and simulate a LAN using Cisco Packet Tracer / GNS3

Aim

To design and simulate a Local Area Network and configure communication among multiple hosts using Cisco Packet Tracer / GNS3.

Objectives

- To build a LAN topology using switches and end devices.
- To assign IP addresses to hosts and verify communication.
- To understand how a switch forwards frames using its MAC address table.

Program / Procedure

Topology: 1 switch (2960) connected to 4 PCs using straight-through cables. Network: 192.168.1.0/24.

1. Place one switch and four PCs in Packet Tracer and connect them with copper straight-through cables.
2. Assign static IPs: PC0 192.168.1.1, PC1 192.168.1.2, PC2 192.168.1.3, PC3 192.168.1.4 with mask 255.255.255.0.
3. Wait until all link lights turn green.
4. From PC0, ping all other PCs and verify replies.
5. On the switch, check the learned MAC addresses.

```
Switch> enable
Switch# show mac address-table
Switch# show ip interface brief
```

Conclusion

A LAN with multiple hosts was designed and simulated. All hosts communicated successfully, and the switch learned MAC addresses to forward frames to the correct port.

Practical No. 4

Static routing between multiple networks

Aim

To configure and verify static routing between multiple networks using routers in Cisco Packet Tracer.

Objectives

- To understand how routers forward packets between different networks.
- To configure static routes manually.
- To verify the routing table and end-to-end connectivity.

Program / Procedure

Topology: PC0 (192.168.1.2/24) – R1 – serial link 10.0.0.0/30 – R2 – PC1 (192.168.2.2/24).

Router R1:

```
Router> enable
Router# configure terminal
Router(config)# hostname R1
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface serial 0/0/0
R1(config-if)# ip address 10.0.0.1 255.255.255.252
R1(config-if)# clock rate 64000
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
R1(config)# end
R1# write memory
```

Router R2:

```
Router(config)# hostname R2
R2(config)# interface gigabitEthernet 0/0
R2(config-if)# ip address 192.168.2.1 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)# interface serial 0/0/0
R2(config-if)# ip address 10.0.0.2 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.1
```

Verification: set the PC gateways to 192.168.1.1 and 192.168.2.1, then run:

```
R1# show ip route
PC0> ping 192.168.2.2
PC0> tracert 192.168.2.2
```

Conclusion

Static routes were configured on both routers and the routing table showed the routes marked "S". PC0 and PC1 on different networks communicated successfully.

Practical No. 5

Dynamic routing protocols – RIP and OSPF

Aim

To configure dynamic routing protocols RIP and OSPF and compare their performance.

Objectives

- To configure RIP version 2 and OSPF on routers.
- To observe automatic route learning and convergence.
- To compare RIP and OSPF based on metric, convergence and scalability.

Program / Procedure

Use the same topology as Practical 4 (remove the static routes first using "no ip route ...").

RIP v2 configuration (on R1; similar on R2 with its own networks):

```
R1(config)# router rip
R1(config-router)# version 2
R1(config-router)# network 192.168.1.0
R1(config-router)# network 10.0.0.0
R1(config-router)# no auto-summary
```

OSPF configuration (on R1; similar on R2):

```
R1(config)# router ospf 1
R1(config-router)# network 192.168.1.0 0.0.0.255 area 0
R1(config-router)# network 10.0.0.0 0.0.0.3 area 0
```

Verification:

```
R1# show ip route
R1# show ip protocols
R1# show ip ospf neighbor
PC0> ping 192.168.2.2
```

Comparison:

Feature	RIP	OSPF
Type	Distance vector	Link state
Metric	Hop count (max 15)	Cost (bandwidth based)
Admin distance	120	110
Convergence	Slow	Fast
Scalability	Small networks	Large networks
Updates	Periodic, full table	Triggered, only changes

Conclusion

RIP and OSPF were configured and routes were learned dynamically. RIP is simple but

slow and limited to small networks, while OSPF converges faster and scales better for large enterprise networks.

Practical No. 6

Layer-2 switching: VLAN, trunking and Inter-VLAN routing

Aim

To configure Layer-2 switching features including VLAN creation, trunking and Inter-VLAN communication.

Objectives

- To create VLANs and assign switch ports to them.
- To configure a trunk link between switch and router.
- To enable communication between VLANs using router-on-a-stick.

Program / Procedure

Topology: PC1 and PC2 (VLAN 10 Sales, 192.168.10.0/24), PC3 and PC4 (VLAN 20 HR, 192.168.20.0/24) on a switch, connected to a router through a trunk port.

Switch configuration:

```
Switch(config)# vlan 10
Switch(config-vlan)# name Sales
Switch(config-vlan)# vlan 20
Switch(config-vlan)# name HR
Switch(config-vlan)# exit
Switch(config)# interface range fastEthernet 0/1-2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config)# interface range fastEthernet 0/3-4
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config)# interface fastEthernet 0/24
Switch(config-if)# switchport mode trunk
```

Router (router-on-a-stick):

```
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# no shutdown
Router(config)# interface gigabitEthernet 0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config)# interface gigabitEthernet 0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
```

Verification:

```
Switch# show vlan brief
Switch# show interfaces trunk
PC1> ping 192.168.20.2
```

Conclusion

VLANs and trunking were configured successfully. Hosts within the same VLAN

communicated directly and, before the router sub-interfaces were configured, were isolated from other VLANs. After configuring Inter-VLAN routing, communication between VLANs worked.

Practical No. 7

Network services – DHCP and DNS

Aim

To configure network services such as DHCP and DNS servers in a simulated environment and verify client connectivity.

Objectives

- To automate IP address assignment using a DHCP server.
- To resolve domain names to IP addresses using a DNS server.
- To verify that clients obtain configuration and resolve names correctly.

Program / Procedure

Topology: 1 switch, 1 server (192.168.1.10), 3 PCs, network 192.168.1.0/24.

1. Assign the static IP 192.168.1.10 / 255.255.255.0 to the server.
2. Server > Services > DHCP: turn service On; set pool name LAN-POOL, default gateway 192.168.1.1, DNS server 192.168.1.10, start IP 192.168.1.50, subnet mask 255.255.255.0, maximum users 50; click Save.
3. Server > Services > DNS: turn service On; add record Name: www.college.com, Type: A Record, Address: 192.168.1.10; click Add.
4. On each PC select Desktop > IP Configuration > DHCP and check that an address is received.
5. On a PC open the command prompt and verify:

```
PC> ipconfig /all
PC> ping www.college.com
PC> nslookup www.college.com
```

Alternative: DHCP on a router:

```
Router(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.10
Router(config)# ip dhcp pool LAN-POOL
Router(dhcp-config)# network 192.168.1.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.1.1
Router(dhcp-config)# dns-server 192.168.1.10
```

Conclusion

DHCP and DNS services were configured. Clients automatically received IP addresses and the domain name was resolved to the server IP, confirming successful operation of both services.

Practical No. 8

Packet capture and analysis using Wireshark

Aim

To capture and analyze network traffic using Wireshark and identify Ethernet, IP, TCP, UDP, ICMP, HTTP and DNS packets.

Objectives

- To capture live packets on a network interface.
- To apply display filters to isolate specific protocols.
- To examine packet headers at different layers.

Program / Procedure

1. Open Wireshark and select the active interface (Wi-Fi / Ethernet), then click Start capture.
2. Generate traffic: run ping, nslookup, and open an HTTP website in the browser.
3. Stop the capture and apply display filters one by one.
4. Select a packet and expand the layers in the details pane to study the headers.

Protocol	Display Filter	What to observe
Ethernet	eth	Source and destination MAC, type field
IP	ip	Source and destination IP, TTL, protocol
TCP	tcp	Ports, sequence numbers, SYN/ACK flags, 3-way handshake
UDP	udp	Source and destination ports, length
ICMP	icmp	Echo request and echo reply (ping)
HTTP	http	GET request, response codes
DNS	dns	Query and response for a domain name

Useful filter examples:

```
ip.addr == 192.168.1.2
tcp.port == 80
dns.qry.name == "example.com"
```

Conclusion

Network traffic was captured and analyzed using Wireshark. Ethernet, IP, TCP, UDP, ICMP, HTTP and DNS packets were identified with display filters, and their header fields were studied.

Practical No. 9

Basic network security: ACL, firewall rules and secure remote access (SSH/Telnet)

Aim

To configure basic network security by implementing Access Control Lists (ACLs), firewall rules and secure remote access using SSH.

Objectives

- To filter traffic using standard and extended ACLs.
- To apply firewall-style permit/deny rules.
- To configure SSH and compare it with Telnet.

Program / Procedure

Use a topology of two LANs connected by a router: LAN1 192.168.1.0/24, LAN2 192.168.2.0/24 with a server at 192.168.2.10.

1. Standard ACL (block host 192.168.1.5 from reaching LAN2):

```
R1(config)# access-list 10 deny host 192.168.1.5
R1(config)# access-list 10 permit any
R1(config)# interface gigabitEthernet 0/1
R1(config-if)# ip access-group 10 out
```

2. Extended ACL / firewall rule (allow only HTTP to the server, deny the rest):

```
R1(config)# ip access-list extended FIREWALL
R1(config-ext-nacl)# permit tcp 192.168.1.0 0.0.0.255 host 192.168.2.10 eq 80
R1(config-ext-nacl)# deny ip any any
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ip access-group FIREWALL in
```

3. Secure remote access using SSH:

```
R1(config)# hostname R1
R1(config)# ip domain-name college.local
R1(config)# crypto key generate rsa modulus 1024
R1(config)# username admin secret Admin@123
R1(config)# ip ssh version 2
R1(config)# line vty 0 4
R1(config-line)# transport input ssh
R1(config-line)# login local
```

Verification:

```
R1# show access-lists
PC> ssh -l admin 192.168.1.1
PC> telnet 192.168.1.1     (should fail after SSH-only is set)
```

Conclusion

ACLs, firewall rules and SSH were configured successfully. Unwanted traffic was

blocked while permitted traffic passed, and remote login became secure because SSH encrypts the session, unlike Telnet which sends data in plain text.

Practical No. 10

Mini Skill Development Project: Secure Enterprise Network (College)

Aim

To design, configure and troubleshoot a secure enterprise network for an organization (college) integrating VLANs, routing, DHCP, DNS and network security.

Objectives

- To plan an IP addressing scheme and VLAN structure for an organization.
- To integrate VLANs, inter-VLAN routing, DHCP and DNS in one network.
- To apply ACLs and SSH for security, and to troubleshoot faults.

Program / Procedure

Scenario: A college network with Admin, Faculty and Lab departments, one server room and a router connected to the internet side.

Department	VLAN	Network	Gateway
Admin	10	192.168.10.0/24	192.168.10.1
Faculty	20	192.168.20.0/24	192.168.20.1
Computer Lab	30	192.168.30.0/24	192.168.30.1
Server Room	40	192.168.40.0/24	192.168.40.1

Steps:

1. Draw the topology in Packet Tracer: core router, switches per department, access switch ports, DHCP/DNS server (192.168.40.10), PCs.
2. Create VLANs 10, 20, 30, 40 and assign access ports; configure the router-facing port as trunk.
3. Configure router sub-interfaces with dot1Q encapsulation for Inter-VLAN routing (as in Practical 6).
4. Configure DHCP pools for each VLAN on the server (or router) and DNS records such as www.college.com.
5. Apply ACLs: allow Admin and Faculty access to the server, block Lab VLAN from Admin VLAN.
6. Configure SSH on router and switches, disable Telnet, set enable secret and console/VTY passwords.
7. Test: DHCP address assignment, ping between VLANs, name resolution, ACL block/permit and SSH login.
8. Troubleshoot using show vlan brief, show ip route, show access-lists and ping/tracert; document faults and fixes.

```
Router(config)# ip access-list extended LAB-POLICY
Router(config-ext-nacl)# deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
```

```
Router(config-ext-nacl)# permit ip any any
Router(config)# interface gigabitEthernet 0/0.30
Router(config-subif)# ip access-group LAB-POLICY in
```

Conclusion

A secure enterprise network for the college was designed, configured and tested. VLAN segmentation, inter-VLAN routing, DHCP, DNS, ACLs and SSH worked together, and troubleshooting tools were used to find and fix connectivity issues.